EN3150 Assignment 03
Section 6: Final Comparison and Trade-off Discussion

Group Draco – *Dealing with AI from the core*

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1 Final Comparison and Trade-off Discussion

The final comparison focuses on Model B, MobileNetV2, and ShuffleNetV2 1.0x. Model A is retained only as a baseline in the Pareto plots. All three final candidates use the same packaged DeFungi split. Test metrics are measured from saved experiments; parameter counts and checkpoint sizes are reproducible from the code/files; MACs are calculated under one common Conv2d+Linear convention for a $3 \times 64 \times 64$ input.

Table 1: Final model comparison.

Model	Test Acc.	Macro F1	Parameters	Size	MACs
Model B	70.43%	0.6927	37,957	0.176 MiB	7.26 M
ShuffleNetV2 1.0x	83.76%	0.8590	1,258,729	4.966 MiB	11.75 M
MobileNetV2	86.31%	0.8805	2,230,277	8.740 MiB	24.45 M

1.1 Accuracy

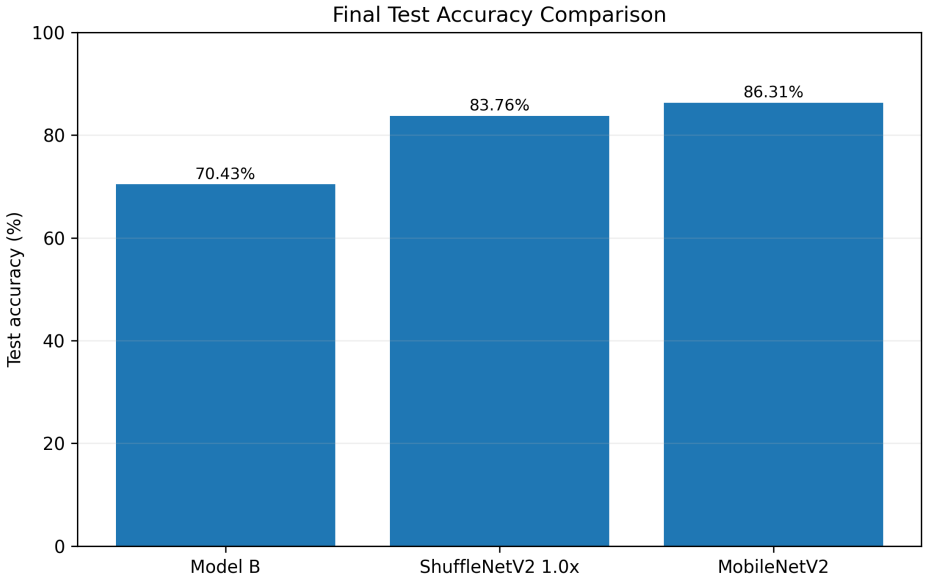
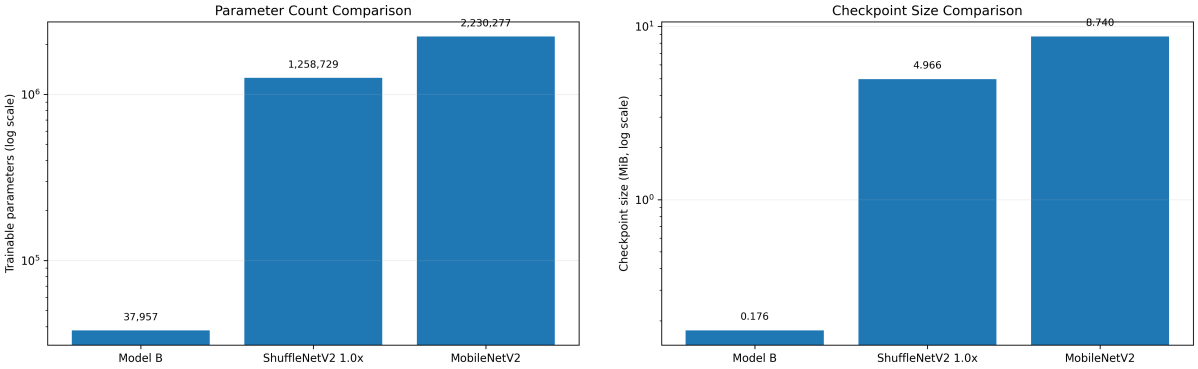


Figure 1: Test accuracy.

MobileNetV2 improves on Model B by 15.88 percentage points, while ShuffleNetV2 improves by 13.33 percentage points. The pre-trained networks also improve class-wise robustness; H2 recall rises from 0.4387 for Model B to approximately 0.66 for ShuffleNetV2 and 0.77 for MobileNetV2.

1.2 Memory Footprint



(a) Parameters.

(b) Checkpoint size.

Figure 2: Memory-related comparisons.

ShuffleNetV2 has 33.16× as many parameters and a 28.23× larger checkpoint than Model B. MobileNetV2 has 58.76× as many parameters and a 49.68× larger checkpoint. These file-size comparisons do not directly equal runtime RAM; peak activation buffers and framework/operator overhead must be measured on the target device.

1.3 Computational Cost

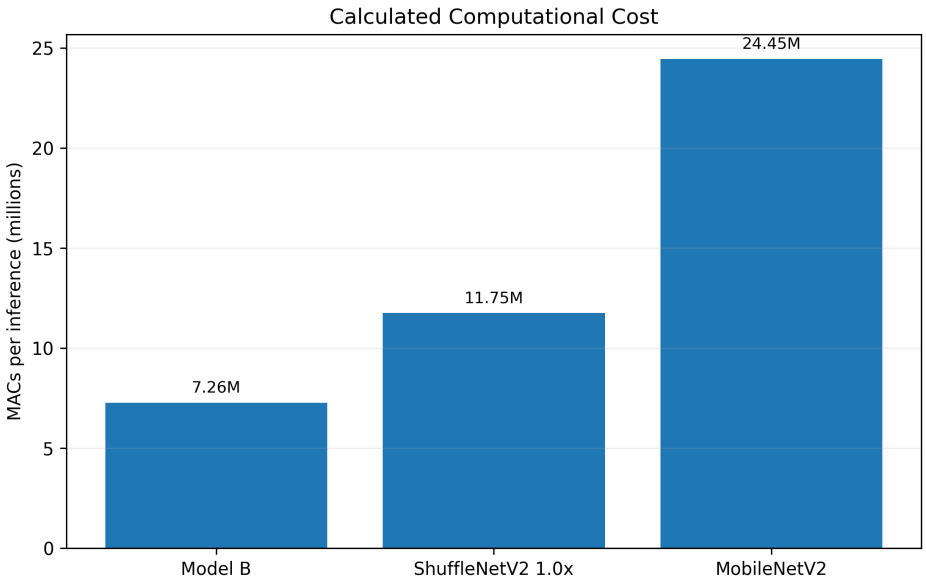
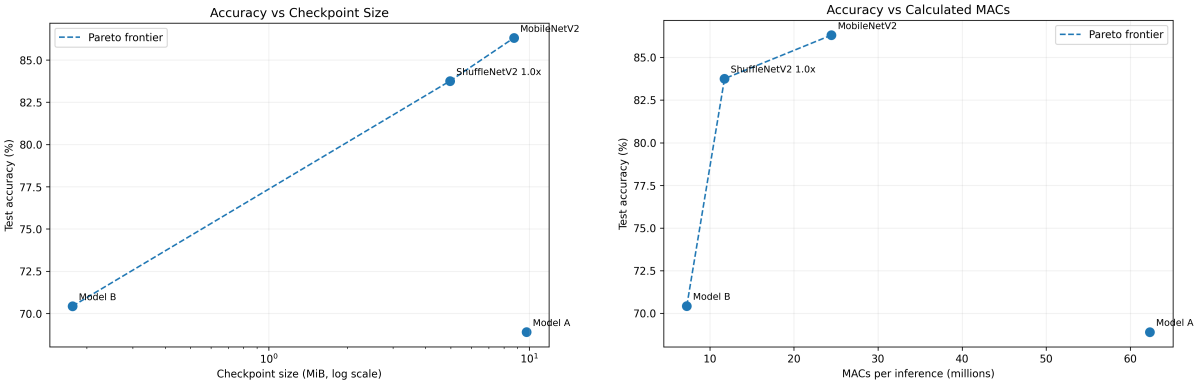


Figure 3: Calculated Conv/Linear MACs per 64×64 image.

Model B requires 7.26 M MACs. ShuffleNetV2 requires 11.75 M (1.62× Model B), while MobileNetV2 requires 24.45 M (3.37× Model B). The project does not contain a valid controlled device-latency measurement, so no latency claim is made and training time is not used as a substitute.

1.4 Accuracy-Efficiency Trade-offs



(a) Accuracy vs size.

(b) Accuracy vs MACs.

Figure 4: Pareto trade-offs including Model A as a baseline.

Model A is dominated by Model B because Model B is both smaller and more accurate. Model B, ShuffleNetV2, and MobileNetV2 form distinct non-dominated operating points: minimum resource cost, balanced accuracy-efficiency, and maximum accuracy respectively.

1.5 Advantages and Limitations

Model B advantages: sub-100k parameters, very small storage footprint, lowest MAC count, simple depthwise-separable blocks, and no dependence on pre-training. Its limitations are lower classification accuracy and capacity, and difficulty on H2.

SOTA advantages: transfer learning provides much stronger accuracy and class-wise performance. MobileNetV2 is the accuracy leader; ShuffleNetV2 is the better efficiency compromise. Their limitations are larger memory and arithmetic requirements, and deployment still depends on peak RAM, quantization, operator support, and measured target-device latency.

1.6 Deployment Recommendation

- **Strictest resource budget: Model B.** It is the smallest model by a large margin. MCU deployment should be confirmed with quantization, peak-RAM profiling, and on-device timing.
- **Balanced choice: ShuffleNetV2 1.0x.** It gains 13.33 percentage points over Model B while requiring substantially fewer resources than MobileNetV2.
- **Accuracy-first choice: MobileNetV2.** It reaches 86.31% test accuracy and 0.8805 macro F1, the best measured classification performance.

2 Conclusion

The experiments demonstrate a clear three-way trade-off. Model B minimizes memory and computational cost; ShuffleNetV2 provides a strong middle ground; MobileNetV2 maximizes accuracy. No single architecture is universally best, so the deployment choice should be tied to the actual memory, latency, and accuracy budget of the target edge platform.